

Study Guide 7: Bonds with Embedded Options: Callable, Puttable and Convertible

Fixed Income Study Guides

Many bonds carry an option that lets the issuer or the investor change the cash flows. To value them you need two ideas: how an option works, and how interest rates move. This guide starts with the options review in Bodie Chapter 2, then applies it to callable, puttable and convertible bonds.

1. Options review (Bodie Chapter 2.5, Derivative Markets)

An **option** gives its buyer the **right, but not the obligation**, to buy or sell an asset at a fixed price (the **strike** or exercise price) on or before a set date. The buyer pays a **premium** for that right; the seller (writer) collects it and must perform if the buyer exercises.

	Call option	Put option
Buyer's right	Buy the asset at the strike	Sell the asset at the strike
Buyer profits when	The asset price rises above the strike	The asset price falls below the strike
Payoff at expiration	$\max(S - K, 0)$	$\max(K - S, 0)$

$$\text{Call payoff} = \max(S_T - K, 0) \quad \text{Put payoff} = \max(K - S_T, 0) \quad \text{Profit} = \text{Payoff} - \text{Premium}$$

Example: a call with strike 100 bought for a premium of 4. If the asset ends at 110, the payoff is 10 and the profit 6. At 95, the call expires worthless and the buyer loses the 4 premium. A put with strike 100 pays 10 if the asset ends at 90.

Money and value

- **In the money:** exercising now would pay (call: $S > K$; put: $S < K$). **At the money:** $S = K$.
- **Out of the money:** exercising now would not pay.
- **Option value = intrinsic value + time value.** Intrinsic value is the payoff from exercising now; time value is what the chance of a better outcome is worth before expiration.
- **Value rises with volatility and time to expiration** for both calls and puts: more uncertainty means more chance of a large payoff, while losses are capped at the premium.

Exercise styles

Style	Can exercise	Bond example
European	Only at expiration	A bond callable on one date only
American	Any time up to expiration	A bond callable continuously after its protection period
Bermudan	On a set of dates	A bond callable on each coupon date (the most common case)

Options vs futures. A futures contract **obligates** both sides to trade at the futures price on the delivery date, so gains and losses are symmetric and no premium is paid up front. An option's

payoff is **asymmetric**: the buyer's loss is limited to the premium. That asymmetry is exactly what an embedded bond option adds.

Bridge to bonds. Replace the stock with a bond and its price with the level of interest rates:

- A **callable** bond: the issuer owns a **call on the bond**. It is in the money when rates fall and the bond's value rises above the call price.
- A **puttable** bond: the investor owns a **put on the bond**. It is in the money when rates rise and the bond's value falls below the put price.
- A **convertible** bond: the investor owns a **call on the issuer's stock**, paid for with the bond.

2. Callable bonds

The issuer may redeem the bond before maturity at a set **call price**, usually par or a small premium that steps down over time.

Key terms

- **Call protection**: a period after issue when the bond cannot be called (for example, "10 non-call 5": 10-year maturity, callable after 5 years).
- **Call schedule**: the dates and prices at which the bond can be called; usually each coupon date after protection ends (Bermudan).
- **Make-whole call**: the call price is the present value of remaining payments at a Treasury yield plus a small spread. It is so expensive that it is used mainly for mergers or restructurings, not for refinancing.
- U.S. Treasuries issued today are not callable; callables are common among corporates, municipalities and agency debt.

Why issuers use them. When rates fall, the issuer can call the bond and refinance at a lower coupon, like a homeowner refinancing a mortgage. The call also allows early retirement of debt after an asset sale or restructuring.

What the investor gives up and gets

- The investor is short the call, so the bond is worth less than an otherwise identical option-free (straight) bond and pays a **higher yield** in compensation.
- **Reinvestment risk**: calls happen when rates are low, so proceeds are reinvested at lower rates.
- **Price compression**: as rates fall, the price stops rising near the call price. This is **negative convexity** (Study Guide 5).

$$V_{callable} = V_{straight} - V_{call}$$

Yield measures: yield to maturity, yield to each call date, and **yield to worst**, the lowest of them.

$$P = \sum_{k=1}^{n_c} \frac{C/2}{(1 + \frac{y_c}{2})^k} + \frac{CP}{(1 + \frac{y_c}{2})^{n_c}}$$

Example (from Study Guide 4): a 10-year 5% bond at 105, callable in 5 years at 101. YTM = 4.377%, YTC = 4.067%, so YTW = **4.067%**. A premium callable is usually priced to its call; a discount callable to its maturity.

3. Puttable bonds

The investor may sell the bond back to the issuer at a set **put price**, usually par, on one or more dates before maturity.

Why they are used

- **Investors** get protection against rising rates and, to a degree, against credit deterioration: if the bond's value falls below the put price, they hand it back and reinvest at the new higher rates.
- **Issuers** pay a **lower coupon** in return, and may attract buyers who would otherwise avoid long maturities.
- The issuer's risk: puts are exercised just when refinancing is most expensive, sometimes when the issuer is in trouble.

$$V_{putable} = V_{straight} + V_{put}$$

Price floor: as rates rise, the price stops falling near the put price, which shortens effective duration.

Yield to put uses the put date and put price in place of maturity and par. Example: a 10-year 5% bond at 97, puttable at 100 in 3 years. YTM = 5.392%; yield to put = **6.110%**. The put only adds to the holder's options, so the YTM is the conservative figure; the yield to put shows the return if the investor exercises.

4. Convertible bonds

The investor may exchange the bond for a set number of the issuer's shares.

Term	Definition	Example (\$1,000 par)
Conversion ratio	Shares received per bond	25 shares
Conversion price	Par ÷ conversion ratio	\$1,000 ÷ 25 = \$40
Conversion value	Share price × conversion ratio	\$36 × 25 = \$900
Straight value	Value as an option-free bond	\$850
Minimum value	The larger of straight value and conversion value	\$900
Market conversion premium	(Price – conversion value) ÷ conversion value	(\$1,000 – \$900) ÷ \$900 = 11.1%

$$V_{convertible} = V_{straight} + V_{call\ on\ stock} \quad \text{Floor} = \max(\text{straight value, conversion value})$$

- **Why issuers use them:** a lower coupon than straight debt, and possible future equity issuance at a price above today's share price.
- **Why investors buy them:** bond-like downside (the straight value floor) with equity upside.
- **Behavior:** with the stock well below the conversion price the convertible trades like a bond ("busted"); well above it, like the stock. Most convertibles are also callable, which lets the issuer force conversion once the stock has risen.

$$V_{callable\ convertible} = V_{straight} + V_{call\ on\ stock} - V_{issuer\ call}$$

5. Pricing bonds with embedded options

A single yield cannot price an option, because the option's value depends on how rates **might** move. The standard tool is a **binomial interest rate tree**: rates branch up or down each period, the bond is valued backward from maturity, and at each exercise date the option holder's choice is applied.

The rule at each exercise node

$$V_{callable} = \min(V_{continue}, CP) \quad V_{putable} = \max(V_{continue}, PP)$$

$$V_{continue} = \frac{\frac{1}{2}(V_{up} + C) + \frac{1}{2}(V_{down} + C)}{1 + r_{node}}$$

Worked example. A 3-year, 5% annual-coupon bond. It is callable (or putable) at 100 only at the end of year 2, after that year's coupon. Up and down moves each have probability $\frac{1}{2}$. The tree is illustrative; in practice it is calibrated so it reprices on-the-run Treasuries exactly.

Date	Rates at each node
Today	3.0%
Year 1	4.5% (up), 2.5% (down)
Year 2	6.0% (up-up), 4.0% (up-down), 2.0% (down-down)

Year 2 (values before the option, ex-coupon): $105 \div (1 + r)$ at each node.

Year 2 node	Rate	Straight	Callable (min with 100)	Putable (max with 100)
Up-up	6.0%	99.057	99.057	100.000 (put)
Up-down	4.0%	100.962	100.000 (called)	100.962
Down-down	2.0%	102.941	100.000 (called)	102.941

Year 1 (ex-coupon), then today:

Node	Straight	Callable	Putable
Year 1 up (4.5%)	100.487	100.027	100.939
Year 1 down (2.5%)	104.343	102.439	104.343
Today (3.0%)	104.286	103.139	104.505

- Value of the issuer's call = $104.286 - 103.139 = \mathbf{1.147}$.
- Value of the investor's put = $104.505 - 104.286 = \mathbf{0.219}$. The put is worth less because only one node puts it in the money.

Effective duration and convexity. Shift every rate in the tree ± 25 bp and reprice (formulas from Study Guide 5):

Bond	Price	Effective duration	Effective convexity
Straight	104.286	2.77	10.5
Callable	103.139	2.10	6.7
Putable	104.505	2.56	9.3

Both options shorten duration: the call caps the price when rates fall, the put floors it when rates rise. The callable's convexity drops; with more of the tree near the call price (lower rates or a finer tree) it turns negative.

Volatility matters. Higher rate volatility makes any option more valuable: a callable's price falls and a puttable's price rises. A straight bond's value does not depend on volatility.

6. Spreads: Z-spread and OAS

- **Z-spread:** the constant spread over the spot (zero) curve that makes the discounted cash flows equal the price, ignoring the option.
- **Option-adjusted spread (OAS):** the constant spread added to every rate in the tree so the model price equals the market price. It is the spread for credit and liquidity with the option removed, which makes callable, puttable and straight bonds comparable.

$$\text{Option cost} = Z\text{-spread} - OAS$$

For a callable the option cost is positive ($OAS < Z\text{-spread}$: part of the extra yield just pays for the call). For a puttable it is negative ($OAS > Z\text{-spread}$).

7. Summary: what the option changes

	Callable	Puttable	Convertible
Who owns the option	Issuer	Investor	Investor (a call on the stock)
Price vs straight bond	Lower	Higher	Higher
Yield vs straight bond	Higher	Lower	Much lower
Yield to quote	Yield to worst	YTM and yield to put	YTM plus conversion premium
Effective duration	Shorter, especially when rates fall	Shorter, especially when rates rise	Depends on the stock; low when equity-like
Convexity	Can turn negative	Positive, enhanced	Positive equity convexity
More volatility	Price falls	Price rises	Price rises
Right valuation tool	Tree or OAS model	Tree or OAS model	Two-factor (rates and stock) or equity option model

8. Tools

TI BA II Plus Professional

- Yield to call: 2nd BOND; set RDT to the call date and RV to the call price (for example 101), enter PRI = 105, then CPT YLD → **4.067**.
- Yield to put: same steps with the put date and put price (RV = 100), PRI = 97 → **6.110**.
- Tree node values: $105 \div 1.06 = \rightarrow 99.057$, and so on backward through the tree.

HP 12C

- Yield to call (the bond functions assume redemption at 100, so use TVM): $10 \text{ n} \cdot 105 \text{ CHS PV} \cdot 2.5 \text{ PMT} \cdot 101 \text{ FV} \cdot i \text{ 2 } \times \rightarrow \mathbf{4.067}$.
- Yield to put: $6 \text{ n} \cdot 97 \text{ CHS PV} \cdot 2.5 \text{ PMT} \cdot 100 \text{ FV} \cdot i \text{ 2 } \times \rightarrow \mathbf{6.110}$.

Excel

Task	Formula	Result
Yield to call	=YIELD(DATE(2026,11,15),DATE(2031,11,15),5%,105,101,2,1)	103.17%
Yield to put	=YIELD(DATE(2026,11,15),DATE(2020,11,15),5%,97,100,2,1)	102.90%
Call payoff	=MAX(S-K,0)	
Conversion value	=share_price*conversion_ratio	\$900
Conversion premium	=(price-conv_value)/conv_value	11.1%

A small tree fits in a worksheet: one column per date, each cell = $\text{MIN}((0.5*(\text{up}+C)+0.5*(\text{down}+C))/(1+r), \text{CP})$ for a callable node (MAX with the put price for a puttable node).

Python: binomial tree (tested; output 104.286, 103.139, 104.505 and durations 2.77, 2.10, 2.56)

```
def tree_price(rates, cpn=5.0, face=100.0, kind="straight", strike=100.0, exercise=(2,), shift=0):
    values = [face + cpn] * (len(rates) + 1)  # cash at maturity
    for t in reversed(range(len(rates))):
        step = []
        for i, r in enumerate(rates[t]):
            v = 0.5 * (values[i] + values[i + 1]) / (1 + r + shift)
            if t in exercise and kind == "callable": v = min(v, strike)
            if t in exercise and kind == "puttable": v = max(v, strike)
            step.append(v + (cpn if t > 0 else 0.0))  # add coupon paid at t
        values = step
    return values[0]

rates = [[0.03], [0.045, 0.025], [0.06, 0.04, 0.02]]
for kind in ("straight", "callable", "puttable"):
    p = tree_price(rates, kind=kind)
    pm = tree_price(rates, kind=kind, shift=-0.0025)
    pp = tree_price(rates, kind=kind, shift=0.0025)
    print(kind, round(p, 3), round((pm - pp) / (2 * p * 0.0025), 2))
```

Python: QuantLib callable bond (tested). A 10-year 5% bond, callable at par on each coupon date after year 5, on a flat 4.5% curve with a Hull-White rate model (mean reversion 3%, volatility 1%). Output: straight 103.991, callable 99.868, so the call is worth about 4.12.

```
import QuantLib as ql
today = ql.Date(15, ql.November, 2026)
ql.Settings.instance().evaluationDate = today
curve = ql.YieldTermStructureHandle(ql.FlatForward(today, 0.045, ql.ActualActual(ql.ActualActual,
                                                                                   ql.Compounded, ql.Semiannual)))
sched = ql.Schedule(today, ql.Date(15, ql.November, 2036), ql.Period(ql.Semiannual), ql.NullCalendar,
                    ql.Unadjusted, ql.Unadjusted, ql.DateGeneration.Backward, False)
dc = ql.ActualActual(ql.ActualActual.Bond, sched)
calls = ql.CallabilitySchedule()
for d in sched:
    if ql.Date(15, ql.November, 2031) <= d < ql.Date(15, ql.November, 2036):
        calls.append(ql.Callability(ql.BondPrice(100.0, ql.BondPrice.Clean), ql.Callability.Call))
callable_bond = ql.CallableFixedRateBond(0, 100.0, sched, [0.05], dc, ql.Following, 100.0, today)
callable_bond.setPricingEngine(ql.TreeCallableFixedRateBondEngine(ql.HullWhite(curve, 0.03, 0.01)))
straight = ql.FixedRateBond(0, 100.0, sched, [0.05], dc)
straight.setPricingEngine(ql.DiscountingBondEngine(curve))
print(straight.cleanPrice(), callable_bond.cleanPrice())
```

Change the volatility from 0.01 to 0.02 and the callable's price falls: the call is worth more. Use `ql.Callability.Put` for a puttable bond.

Julia (same logic as the tested Python tree; not run here)

```
function tree_price(rates; cpn = 5.0, face = 100.0, kind = :straight, strike = 100.0, exercise = :call)
    values = fill(face + cpn, length(rates) + 1)
    for t in length(rates):-1:1
        step = Float64[]
        for (i, r) in enumerate(rates[t])
            v = 0.5 * (values[i] + values[i + 1]) / (1 + r + shift)
            t in exercise && kind == :callable && (v = min(v, strike))
            t in exercise && kind == :puttable && (v = max(v, strike))
            push!(step, v + (t > 1 ? cpn : 0.0))
        end
        values = step
    end
    values[1]
end
```

```
rates = [[0.03], [0.045, 0.025], [0.06, 0.04, 0.02]]
tree_price(rates; kind = :callable) # 103.139
```

Julia arrays start at 1, so year 2 is index 3 in exercise.

9. Practice set

1. You buy a call with strike 50 for a premium of 3. The stock ends at 58. Payoff and profit?
Payoff 8, profit 5.
2. A 10-year 6% semiannual bond trades at 103 and is callable at par in 5 years. Find YTM, YTC and YTW. *5.604%, 5.309%; YTW = 5.309%.*
3. Rework the tree example with a call price of 101. *Callable value 103.827; the higher call price makes the call less valuable.*
4. A straight bond is worth 102.50 and the otherwise identical callable 99.80. Value of the call? *2.70.*
5. A convertible has conversion ratio 20, the stock trades at \$55, the straight value is \$950 and the market price is \$1,150. Find conversion value, minimum value and conversion premium. *\$1,100; \$1,100; 4.5%.*
6. A bond's Z-spread is 150 bp and its OAS 110 bp. What is the option cost, and is the bond more likely callable or puttable? *40 bp; callable.*

10. CFA-style review questions

Original questions in the CFA Level I format (three choices, one correct). Not CFA Institute material.

1. The investor in a callable bond is effectively:
 - A. long a call option on the bond
 - B. short a call option on the bond
 - C. long a put option on the bond
2. If interest rate volatility increases, the value of a callable bond most likely:
 - A. increases
 - B. decreases
 - C. is unchanged
3. A straight bond is worth 101.40 and the embedded put is worth 1.20. The puttable bond's value is:
 - A. 100.20

- B. 101.40
 - C. 102.60
4. As interest rates fall well below its coupon rate, a callable bond's price will most likely:
 - A. rise without limit, like a straight bond
 - B. rise more slowly and flatten near the call price
 - C. fall
 5. A convertible bond has conversion ratio 40 and a straight value of \$820; the stock trades at \$22. Its minimum value is:
 - A. \$820
 - B. \$880
 - C. \$1,000
 6. A bond has a Z-spread of 120 bp and an OAS of 90 bp. It is most likely:
 - A. callable
 - B. putable
 - C. option-free
 7. When rates are low, a callable bond's effective duration compared with its modified duration is:
 - A. lower
 - B. the same
 - C. higher

Answer key

1. B. The issuer owns the call; the investor has sold it.
2. B. A more valuable call is subtracted from the straight value.
3. C. $101.40 + 1.20 = 102.60$.
4. B. Negative convexity: the call caps the price.
5. B. Conversion value $40 \times \$22 = \880 exceeds the straight value.
6. A. A positive option cost (30 bp) means the investor is short an option.
7. A. Modified duration ignores the likely call, which shortens the bond's effective life.

11. Bloomberg Terminal exercises (draft: verify on the campus Terminal)

Command	What to do	Check against this guide	Verified
OMON <GO>, OV <GO>	Equity option monitor and option valuation: view calls and puts, strikes, premiums	Section 1: payoffs, moneyness, volatility	☐
SRCH <GO>	Find corporate bonds that are callable, putable or convertible (use the call, put and convertible filters)	Sections 2-4	☐
DES <GO>	Read the call or put schedule, call protection and make-whole terms	Sections 2 and 3	☐
YAS <GO>	Read YTM, yield to call, yield to worst and the workout date	Section 2: YTW	☐

Command	What to do	Check against this guide	Verified
OAS1 <G0>	Read OAS, option value, effective duration and convexity; change the volatility input and watch the price	Sections 5 and 6	☐
OVCV <G0> (confirm)	Convertible valuation: conversion value, premium, bond floor	Section 4	☐

Confirm Excel field names for yield to worst, OAS and conversion premium with FLDS <G0>.

Exercise: pick a callable corporate bond. In OAS1, record the OAS and option value at the default volatility, then at volatility 5 points higher. Explain the change in price using section 5.

12. Reading in Bodie, Kane & Marcus, *Investments*, 13th ed. (2024)

Section	Topic in the book	Supports
2.5 Derivative Markets	Options; Futures Contracts	Section 1: the options review
20.1 The Option Contract	Options trading; American and European options	Section 1
20.2 Values of Options at Expiration	Call and put payoffs	Section 1: payoff formulas
20.5 Option-like Securities	Callable Bonds; Convertible Securities	Sections 2 and 4
14.1 Bond Characteristics	Corporate Bonds: call provisions, convertible bonds, puttable bonds	Sections 2-4
14.3 Bond Yields	Yield to Call	Section 2: YTC and YTW
16.2 Convexity	Duration and Convexity of Callable Bonds	Sections 5 and 7: negative convexity
Chapter 21 Option Valuation	Binomial option pricing	Section 5: backward induction in a tree